

Precalc Lesson 1.11 Homework

Name _____

1. Let $f(x) = \begin{cases} x^2 + 3x - 9 & x \leq -4 \\ 2x + 1 & -4 < x \leq 1 \\ -5x + 7 & x > 1 \end{cases}$

Find $f(-8)$, $f(-4)$, and $f(3)$.

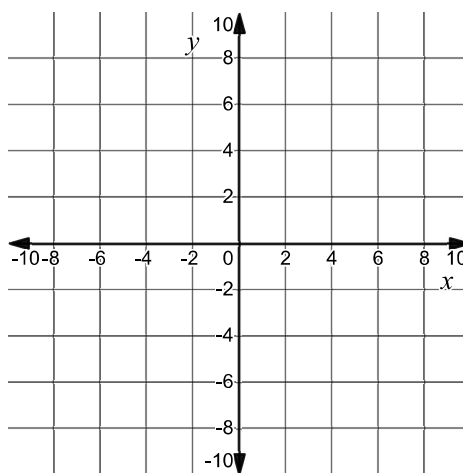
2. The cost, C , in dollars, of shipping a package that weighs w pounds can be modeled by the piecewise function given by

$$C(w) = \begin{cases} 11 & 0 < w < 2 \\ 9 + 1.49w & 2 \leq w < 10 \\ 6 + 1.99w & w \geq 10 \end{cases}$$

- How much does it cost to ship a package that weighs 1.5 pounds?
- How much does it cost to ship a package that weighs 5 pounds?
- How much more does it cost to ship a package that weighs 10 pounds than a package that weighs 9 pounds?

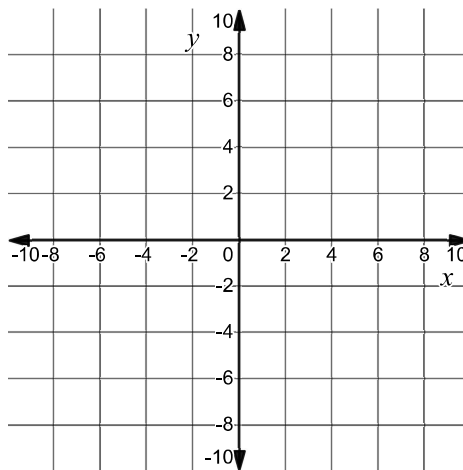
3. Graph the piecewise function given by

$$g(x) = \begin{cases} 4 & -10 < x \leq -5 \\ x & -5 < x \leq 2 \\ -2 & x > 2 \end{cases}.$$



4. Graph the piecewise function given by

$$g(x) = \begin{cases} x + 9 & -6 \leq x < 0 \\ \frac{1}{2}x^2 & 0 \leq x \leq 4 \\ -2x + 11 & 4 < x \leq 8 \end{cases}.$$



5. Walgreens has the following pricing for 4x6 photo prints:
- Quantity of 1-74 costs \$0.37 each.
 - Quantity of 75+ costs \$0.25 each.
- a. Write a piecewise function $C(p)$, that gives the cost of purchasing p prints.
- b. Patty ordered 70 prints, then realized she was still short and ordered 10 more the next day. How much did this mistake cost her? Explain.

6. Let $f(x) = \begin{cases} 2 + 13x & x \leq -3 \\ 25 - x^2 & -3 < x \end{cases}$.

The graph of f can be drawn with an open circle at $(-3, k)$. Find the value of k .

7. Santiago's weekly income after working h hours can be described by the function f below.

$$f(h) = \begin{cases} 13.5h & 0 < h \leq 40 \\ 20.25h - 270 & h > 40 \end{cases}$$

Which of the following conclusions is correct?

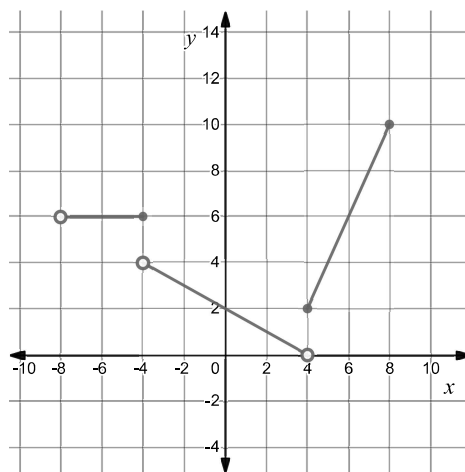
- A) Santiago gets paid \$13.50 an hour for up to 40 hours, and an additional \$20.25 per hour for hours beyond 40.
- B) Santiago has to pay \$270 in taxes if he works more than 40 hours in a week.
- C) Santiago gets paid 1.5 times his hourly rate for hours he works beyond 40 hours.
- D) If Santiago works 43 hours, he gets paid \$870.75.

8. Let $f(x) = \begin{cases} (x+3)^2 & x < -2 \\ 7 & x = -2 \\ \sqrt{3x+15} & x > -2 \end{cases}$.

Which of the following is true about f ?

- A) $f(-2) = 1$
- B) The graph of $f(x)$ has an open circle at $(-2, 3)$.
- C) $f(-6)$ is undefined.
- D) $f(x)$ has three distinct outputs at $x = -2$.

9. The graph of a piecewise function f is shown in the xy -plane for $-8 < x \leq 8$. Write the equation for f .



10. The graph of a piecewise function g is shown in the xy -plane for $-7 \leq x < 6$. Write the equation for g .

